

Prepared by Tomas Torres – Meta AI assist Confidential Brief • June 2024

Zero-Combustion AI

Magnetic-Assisted Power + Waterless Cooling

A pilot-ready plan for Texas data centers

The Problem

448 TWh

Global DC energy last year • 208M tons CO₂ • 1.2T gallons water

945 TWh

Projected by 2030 — AI workloads more than double demand

Texas Grid

Peak strain, interconnection delays, and aquifer stress rise

Core Insight

No perpetual motion. Magnets don't create energy, only **loss**. The real leverage point is **cooling**. Removing **60%** of a data center's parasitic load.

Solution Pillars



Power

Permanent-magnet direct-drive wind/solar + magnetic-geothermal option for base load. Zero on-site combustion.



Cooling

Direct-to-chip two-phase dielectric (PG25) then full immersion. **60%** cooling energy cut vs. air.



Architecture

Designed for 50–100kW racks. Heat reuse for building
time PUE/WUE telemetry.

Pilot Plan

Phase 1

2 racks, 80–150kW total load

Budget: \$250k (cooling skids, retrofit, sensors)

Metrics: PUE, WUE, kWh saved, water gallons

Phase 2

Expand to 1MW pod in Carrollton

Add flywheel + behind-meter solar

Target: PUE ≤ 1.08 , WUE = 0, payback <3yr

Why Now

Vendors are already shipping: **Vertiv CoolPhase**
Uniflair XCA. Industry is moving from air to liquid

restrictions make zero-water designs a competitive

Prepared by Tomas Torres – Meta AI assist v1.0 • June 4, 2026

5-Slide Pitch Outline

Zero-Combustion AI Data Center — Investor / Partner

Problem

448 TWh global DC use (2025) = 208M tCO₂e

945 TWh projected by 2030

Texas: grid volatility, ERCOT peaks, water s

Result: higher opex, permitting risk, commu

Physics Reality

No "free energy" — magnets enable ultra-ef

Permanent-magnet generators: >96% effici

Magnetic bearings: near-zero friction for flyw

Cooling is 30–40% of load — that's the leve

Power Design

Primary: grid + behind-meter PM wind/solar

Ride-through: magnetic-bearing flywheels (r

Base load option: SMR or geothermal — ze

Outcome: resilient, clean, < \$0.06/kWh blen

Cooling Design

Step 1: Direct-to-chip two-phase (PG25 die)

Step 2: Full immersion for GPU clusters

Closed loop dry coolers — WUE ~0

PUE 1.05–1.1, 30–60% cooling energy reduction

Pilot & ROI

Phase 1: 2 racks, \$250k, 6 months in Carro

Phase 2: Scale to 1MW pod, add flywheel +

ROI drivers: energy savings, water avoided,

Ask: site partner, \$250k pilot funding, vendo